THE MASTERMIND COACHING CLASSES

CBSE / RBSE · Nursery – Class XII · Road No. 5, VKIA, Jaipur · 9887258679

CLASS X — SCIENCE (086) · FULL SYLLABUS PRACTICE PAPER

Session 2026–27 | CBSE / RBSE Pattern | Section A Biology · Section B Chemistry · Section C Physics

Class: 10	Subject: Science	Set 01	Time: 3 Hours	Max. Marks: 80

General Instructions:

1. This question paper consists of 39 questions in 3 sections. Section A is Biology, Section B is Chemistry and Section C is Physics.
2. All sections are compulsory. However, internal choice is provided in some questions. A student is expected to attempt only one of these questions.
3. Draw neat, labelled diagrams wherever required. Use of calculator is not allowed.

SECTION A — BIOLOGY — 16 questions · 30 marks

Q1–Q7 MCQs. Q8–Q9: Assertion (A) / Reason (R) – choose A, B, C or D.

1. A leaf of a potted plant is kept in the dark for 24 hours. A part of it is then covered with black paper, the plant is kept in sunlight for some hours and the leaf is tested with iodine solution. Only the uncovered part turns blue-black. The experiment shows that: **1**

A. chlorophyll is necessary for photosynthesis

B. light is necessary for photosynthesis

C. carbon dioxide is necessary for photosynthesis

D. oxygen is released in photosynthesis

2. A person touches a hot plate and pulls his hand away before he feels the pain. The correct path of the impulse in this action is: **1**

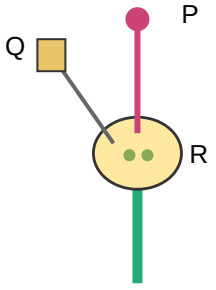
A. receptor → sensory neuron → spinal cord → motor neuron → effector

B. receptor → motor neuron → spinal cord → sensory neuron → effector

C. receptor → sensory neuron → brain → motor neuron → effector

D. effector → sensory neuron → spinal cord → motor neuron → receptor

3. The figure shows a longitudinal section of a flower. Pollen grains germinate on the part labelled P. The male gamete finally fuses with the female gamete present in the part labelled: **1**



Longitudinal section of a flower (simplified)

A. P B. Q C. R D. None of these

4. A pea plant with round, yellow seeds (RrYy) is self-pollinated. The fraction of the offspring expected to have wrinkled, green seeds is: **1**

A. 9/16 B. 3/16 C. 1/16 D. 4/16

5. In a food chain grass → grasshopper → frog → snake → eagle, 10 000 J of energy is available at the grass level. According to the 10% law, the energy available to the eagle is: **1**

A. 1 J B. 10 J C. 100 J D. 1000 J

6. Transpiration in plants is useful because it: **1**

A. makes the plant lose all its water so that it can absorb fresh water

B. creates a pull that helps in upward movement of water and minerals and also helps in cooling the plant

C. transports the food made in the leaves to the roots

D. closes the stomata and stops photosynthesis

7. A school compost pit receives (i) vegetable peels, (ii) polythene bags, (iii) dry leaves and (iv) broken glass. Which of these will be broken down by decomposers? **1**

A. (i) and (iii) only

B. (i), (ii) and (iii)

C. (ii) and (iv) only

D. all four

8. **Assertion (A):** The sex of a child is determined by the father and not by the mother. **1**
Reason (R): The father produces two types of gametes, carrying X and Y chromosomes, whereas the mother produces only X-type gametes.

A. Both A and R are true, and R is the correct explanation of A.

B. Both A and R are true, but R is not the correct explanation of A.

C. A is true but R is false.

D. A is false but R is true.

9. **Assertion (A):** Sexual reproduction gives rise to variation among the offspring. **1**
Reason (R): In sexual reproduction the offspring receive a different combination of genes from two parents and errors may occur during DNA copying.

A. Both A and R are true, and R is the correct explanation of A.

B. Both A and R are true, but R is not the correct explanation of A.

C. A is true but R is false.

D. A is false but R is true.

10. Why is the rate of breathing in aquatic organisms much faster than in terrestrial organisms? **2**

11. State any two advantages of vegetative propagation. **2**

OR

Write any two differences between sexual and asexual reproduction.

12. Why is the use of iodised salt recommended in our diet? Name the hormone whose synthesis needs iodine and the disease caused by its deficiency. **2**

13. Name the hormone secreted by the pancreas that controls blood sugar. What will happen if its secretion decreases? State one function of adrenaline. **3**

14. A tall pea plant (TT) is crossed with a dwarf pea plant (tt). Show the cross up to the F_2 generation using a Punnett square and write the phenotypic and genotypic ratios of F_2. **3**

15. A town produces large quantities of garbage every day: kitchen waste, plastic bottles, paper and old refrigerators. The municipal body wants to segregate the waste. Scientists also warn that the use of certain chemicals in refrigerators and aerosols has thinned the protective ozone layer high in the atmosphere.

(a) What is meant by a biodegradable substance? Name one such item from the waste listed above. 1

(b) Why are plastic bottles harmful for the environment? 1

(c) Name the chemical responsible for ozone depletion and state one harmful effect of the depletion on human health. 2

OR

(d) In a food chain, why does the number of organisms decrease at higher trophic levels? State the 10% law.

16. (a) Describe the main events of sexual reproduction in a human female: name the organ that produces the egg, the site of fertilisation, the site of implantation and the role of the placenta. (b) Name any two methods of contraception and state one disadvantage of each type of method you name. **5**

OR

(i) Describe how carbohydrates, proteins and fats are digested in the human alimentary canal, naming the enzyme/juice at each stage. (ii) State the role of villi in the small intestine.

SECTION B — CHEMISTRY — 12 questions · 25 marks

Q17–Q21 MCQs. Q22: Assertion (A) / Reason (R) – choose A, B, C or D.

17. A student burns a strip of magnesium ribbon in air and collects the white ash formed. The ash is dissolved in water and the solution is tested with red and blue litmus paper. Which statement is correct? **1**

A. The ash is MgO and it turns red litmus blue

B. The ash is MgO and it turns blue litmus red

C. The ash is $Mg(OH)_2$ and it turns blue litmus red

D. The ash is $MgCO_3$ and it does not change either litmus

18. Which of the following reactions is both a combination reaction and exothermic in nature? **1**

A. $CaCO_3 \rightarrow CaO + CO_2$ B. $CaO + H_2O \rightarrow Ca(OH)_2$

C. $2H_2O \rightarrow 2H_2 + O_2$ D. $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$

19. Rahul tests four solutions P, Q, R and S with universal indicator and records pH values 2, 7, 10 and 13 respectively. Which solution is most suitable as a mild antacid for heartburn? **1**

A. P B. Q C. R D. S

20. Four metals W, X, Y and Z were tested. W reacts with cold water; X reacts only with steam; Y reacts with dilute HCl but not with water or steam; Z reacts with none of these. The correct decreasing order of reactivity is: **1**

A. Z > Y > X > W B. W > X > Y > Z C. X > W > Y > Z D. W > Y > X > Z

21. The number of covalent bonds (shared pairs of electrons) present in one molecule of ethyne, C_2H_2, is: **1**

A. 3 B. 4 C. 5 D. 6

22. **Assertion (A):** Sodium hydrogencarbonate is used in baking powder. **1**
Reason (R): On heating, it releases carbon dioxide, which makes the cake or bread rise and become soft.

A. Both A and R are true, and R is the correct explanation of A.

B. Both A and R are true, but R is not the correct explanation of A.

C. A is true but R is false.

D. A is false but R is true.

23. Balance the equation $Fe_2O_3 + Al \rightarrow Al_2O_3 + Fe$ and name the type of reaction. State one use of this reaction. **2**

24. Dry HCl gas does not change the colour of dry blue litmus paper but hydrochloric acid solution does. Why? **2**

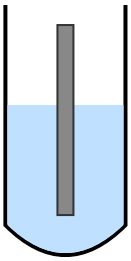
25. Zinc is obtained from its sulphide ore ZnS. Name the steps involved and write the chemical equation for each step. **3**

OR

State three physical properties of ionic compounds and give the reason for each.

26. Draw the structures of the two isomers of butane (C_4H_{10}) and name them. Also write the IUPAC name of $CH_3CH_2CH_2OH$. **3**

27. Ramesh puts a few granules of zinc in a test tube containing dilute sulphuric acid. He observes bubbles of a colourless gas and, when a burning matchstick is brought near the mouth of the test tube, the gas burns with a ‘pop’ sound.



Zinc granules in dilute H_2SO_4

(a) Name the gas evolved and state the test confirming it. 1

(b) Write the balanced chemical equation for the reaction. 1

(c) Would the same observation be made if copper turnings were used in place of zinc? Explain using the reactivity series. 2

OR

(d) What will you observe if the zinc granules are added to copper sulphate solution? Write the equation.

28. (a) Write the chemical name and formula of bleaching powder. How is it prepared? Write the equation. **5**
(b) State two uses. (c) What is observed when carbon dioxide is passed through lime water first in small and then in excess quantity? Write equations.

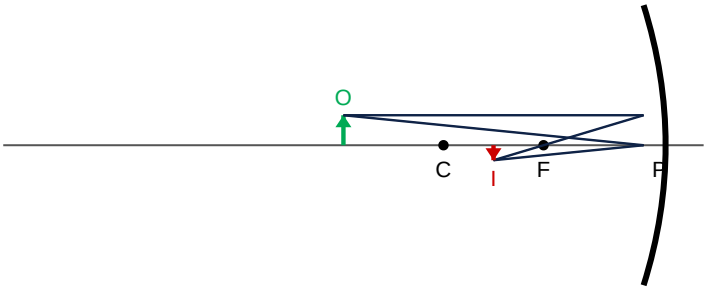
OR

(a) Ethanol reacts with sodium and with ethanoic acid. Write the equation and name the products in each case. (b) How can ethanol be converted into ethene? Write the conditions. (c) Why does ethanol not turn litmus red although it has an –OH group?

SECTION C — PHYSICS — 11 questions · 25 marks

Q29–Q32 MCQs. Q33: Assertion (A) / Reason (R) – choose A, B, C or D.

29. A concave mirror of focal length 10 cm forms an image of an object placed 30 cm from it, as shown in the figure. The image is: **1**



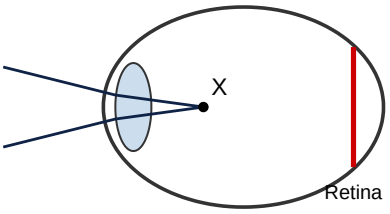
A. real, inverted, diminished and between F and C

B. virtual, erect and magnified

C. real, inverted and of the same size at C

D. real, inverted and magnified beyond C

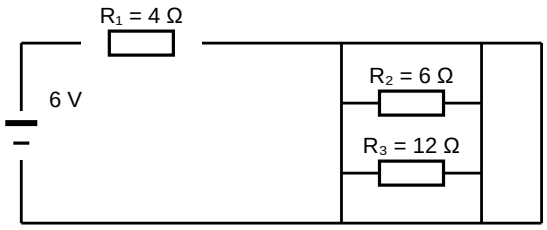
30. A student cannot read the writing on the blackboard from the last bench but can read his book comfortably. The defect of vision and the lens needed to correct it are: **1**



Parallel rays focus at X, in front of the retina

A. myopia; concave lens
B. myopia; convex lens
C. hypermetropia; concave lens
D. hypermetropia; convex lens

31. Three resistors are connected to a 6 V battery as shown in the figure. The current drawn from the battery is: **1**



A. 0.5 A
B. 0.75 A
C. 1.0 A
D. 1.5 A

32. A straight conductor carries current from left to right in the plane of the paper. A uniform magnetic field acts into the plane of the paper. The force on the conductor is directed: **1**

A. upwards, in the plane of the paper
B. downwards, in the plane of the paper
C. into the plane of the paper
D. out of the plane of the paper

33. **Assertion (A):** A magnetic compass needle placed near a wire carrying current gets deflected. **1**
Reason (R): A current-carrying conductor produces a magnetic field around it.

A. Both A and R are true, and R is the correct explanation of A.
B. Both A and R are true, but R is not the correct explanation of A.
C. A is true but R is false.
D. A is false but R is true.

34. A concave mirror has a focal length of 15 cm. An object is placed 10 cm in front of it. Find the position of the image and state its nature. **2**

OR

Define the principal focus of a concave mirror. If the radius of curvature of a concave mirror is 24 cm, what is its focal length (with sign)?

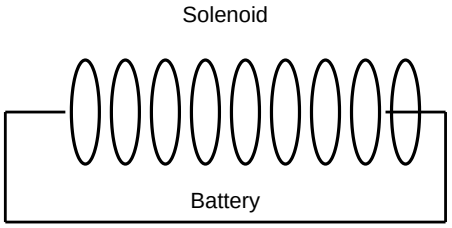
35. A myopic person has a far point at 80 cm. Calculate the focal length and power of the lens needed to correct the defect. State the nature of the lens. **3**

36. Two resistors of 6 Ω and 3 Ω are connected in parallel across a 12 V battery. Calculate (i) the equivalent resistance, (ii) the total current drawn and (iii) the heat produced in the 6 Ω resistor in 10 s. **3**

37. State Snell's law of refraction. The speed of light in a transparent medium is 2×10^8 m/s; find its refractive index (speed of light in air = 3×10^8 m/s). What is the nature of the emergent ray when light passes through a rectangular glass slab? **3**

38. Ananya makes an electromagnet by winding insulated copper wire on a cardboard tube (a solenoid) and connecting it to a battery. She finds that the solenoid behaves like a bar magnet. She also

pushes a bar magnet in and out of a coil connected to a galvanometer and notices that the galvanometer needle deflects.



(a) What is observed in the galvanometer when the magnet is pushed in and pulled out? Name the phenomenon. 1

(b) Write any two ways of increasing the strength of the magnetic field of the solenoid. 1

(c) Draw the pattern of the magnetic field lines of the solenoid. Which end of the solenoid will act as the north pole? 2

OR

(d) State Fleming's right-hand rule. Name a device based on it.

39. An object 5 cm high is placed 30 cm from a convex lens of focal length 20 cm. Find the position, nature and size of the image formed. Draw a ray diagram for the situation. **5**

OR

(a) With a labelled diagram, explain the principle, construction and working of a simple electric motor.
(b) What is the function of the split ring (commutator)?